Eric Medoua

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US Citizen | *Open for Relocation*

SUMMARY

Oracle Certified Java SE 17 Backend/Full Stack Developer with **3 years** of hands-on experience delivering innovative software solutions. Demonstrates expertise in creating efficient, high-performance solutions using **Java, Python, C++, JavaScript, and TypeScript.** Skilled in both **back-end** and **front-end development**, with a focus on crafting resilient **microservices** using **Spring Boot**, **Node.js,** and **Express**, and building dynamic user interfaces with **React, Angular**, and **Vue.js.** Proficient in employing **HTML5, CSS3,** and **Bootstrap** for seamless user interfaces and delivering secure, feature-rich applications with **RESTful APIs.** Experienced in cloud technologies, deploying applications on **AWS (ECS, Fargate, S3, CloudFront), Azure,** and **Google Cloud**, ensuring optimal performance, security, and reliability.

Adept at managing databases such as **MySQL, PostgreSQL, MongoDB, Amazon RDS, DynamoDB,** and **Firebase**, and optimizing them for complex data management needs. Proficient in **DevOps practices**, utilizing tools like **Jenkins, Docker, Kubernetes**, and **Terraform to streamline CI/CD** processes and ensure seamless deployment. Recognized for delivering high-quality solutions by applying advanced technologies such as **Kubernetes**, **EKS, AWS CodePipeline**, and **CloudWatch**.

TECHNICAL SKILLS AND CERTIFICATIONS

- **Certifications:** Oracle Certified Professional Java SE 17 Developer, AWS Certified Solutions Architect - Associate
- **Programming Languages:** Java, JavaScript (ES6), Python, TypeScript, CSS, HTML5, SQL, C++
- **Frameworks & Libraries:** Spring Boot, Spring MVC, Spring Security, Hibernate, JUnit 4/5, JSP/Servlets, React 16, Redux, Node.js, Express.js, Mongoose, Next.js, Angular, Vue.js, jQuery, Bootstrap 4/5, Material UI, Swagger, Flexbox
- **Microservices & Web Services:** RESTful APIs, Spring Cloud, API Gateway
- **CI/CD Tools:** ***Jenkins***
- **DevOps Tools:** ***Ansible, Docker, Kubernetes, Terraform, Git Ops, Kafka***
- **Database:** MySQL, Oracle 11g, MongoDB, PostgreSQL, DynamoDB, Aurora, Amazon Redshift, Elasticsearch, RDS, Firebase
- **Build Tools and Version Control:** Maven, Git, GitHub, Webpack, Babel, Jira
- **Development Tools:** IntelliJ IDEA, Eclipse, Postman, Vite
- **Platforms:** Windows XP/Vista/7/8/10/11, Linux, Unix
- **Cloud Technologies:** AWS (EC2, S3, RDS, Route 53, IAM, VPC, CodePipeline, CodeBuild, CloudWatch, Fargate, SNS), Azure, Google Cloud Platform
- **Other Skills:** JWT Authentication, Stripe API, Axios, Linux/Unix, TCP/IP Protocols, WAN/LAN Networking.

PROFESSIONAL EXPERIENCE *(SELECTED PROJECTS)*

Java Full Stack Developer | ***SynergisticIT*** *– Fremont, CA* | **July, 2022 - Present**

Secure Banking Web Application **April 2024 – Present**

Overview:
Developed a comprehensive banking web platform using the **Java ecosystem**, integrating **Spring MVC** and **Hibernate** for backend functionality and **JSP** for a dynamic frontend. The application enables customers to view account summaries, perform withdrawals, deposits, and transfers, while staff can manage customer accounts and address inquiries. Administrators oversee operations, including managing customers, staff, and branches. The project incorporates **Spring Security** for robust authentication and authorization, ensuring data privacy and secure transactions. Data management leverages **MySQL** integrated via **AWS RDS**, while **AWS S3** handles efficient storage and retrieval of customer profile photos. Deployed on **Tomcat 9**, the application ensures scalability and reliability. Performance optimization is supported by **AWS services**, and the development lifecycle is streamlined with **Maven** for build automation.

Backend Development:

- Designed and implemented the application architecture using **Spring MVC**.
- Developed RESTful APIs to support core banking functionalities and validated them using **Postman**.
- Secured application access with role-based authentication and authorization using **Spring Security**.
- Conducted unit and regression testing with **JUnit** and **Mockito**, ensuring high-quality code.
- Managed customer profile photo storage and retrieval using **AWS S3**, optimizing resource handling.
- Integrated the application with a database via an **AWS RDS instance endpoint**, ensuring efficient data management.

Frontend Development:

- Created dynamic and responsive web pages using **HTML5**, **CSS**, and **Bootstrap 5** for a polished user interface.
- Developed interactive frontend components using **JSP**, integrating REST APIs for backend communication.
- Enhanced user experience with **JavaScript**, providing responsive and intuitive interfaces.

Deployment and CI/CD:

- Deployed the application on **Tomcat 9**, ensuring reliable and secure hosting.
- Utilized **AWS services** for storage, database management, and performance optimization.
- Streamlined the development lifecycle with **Maven**, automating build and dependency management.

Technologies Used: Java, Spring MVC, Hibernate, Spring Security, MySQL, JUnit, Mockito, JavaScript, HTML5, Bootstrap 5, JSP, Maven, AWS S3, AWS RDS, Tomcat 9.

Java Full Stack Developer | ***SynergisticIT*** *– Fremont, CA*

Hotel Reservation Gateway **Sep 2023 – Mar 2024**

Overview:
Developed an advanced web-based hotel reservation system designed for seamless hotel searches, reservations, and feedback management. Leveraging **Java SE 17**, **Spring Boot**, and **Hibernate**, the project features a microservices architecture for scalability and security, with **Spring Security**, and **Spring Framework** for backend functionality, and **Bootstrap** and **HTML5** for a dynamic frontend. The platform enables customers to browse hotels, book rooms, evaluate services, and view their reservation history, while hotel managers efficiently monitor guest feedback and respond to inquiries. The **microservices architecture** ensures modularity and scalability, enabling independent service management. Deployed using **AWS services** for scalable performance, the system ensures seamless operations, robust security, and user satisfaction.

Backend Development:

- Designed and implemented **RESTful microservices** using **Spring Boot**, ensuring modularity and scalability.
- Deployed microservices on **AWS EC2**, integrating **S3** for file storage and **SNS** for notification services.
- Utilized **Spring Security** for implementing user authentication and role-based access control.
- Conducted unit testing with **JUnit 4**, achieving **85% code** coverage to ensure robust functionality.
- Integrated **Spring Data Actuator** to monitor and provide service health metrics for production maintenance.
- Designed and implemented relational databases using **MySQL**, supporting backend infrastructure with efficient data management.

Frontend Development:

- Developed responsive and interactive UI components using **Bootstrap**, **HTML5**, and **CSS** for a visually appealing user experience.

- Incorporated **jQuery** and **Ajax** for dynamic interaction with **backend RESTful web services**, improving responsiveness.
- Ensured form and input validation using **JavaScript** and **Bootstrap**, enhancing usability and security.

Deployment and Version Control:

- Utilized **AWS IAM** for secure access management and **AWS CodeDeploy** for automated deployment processes.
- Managed version control and collaboration through **GitHub**, ensuring efficient team workflows.
- Deployed the application on **Tomcat 9**, ensuring reliable and secure hosting.

Technologies Used: Java, Spring 5, Spring Security, JUnit 4, MySQL, HTML5, CSS, Bootstrap, JSP, jQuery 3, Ajax, Maven, AWS EC2, AWS S3, AWS SNS, AWS IAM, AWS CodeDeploy, GitHub, Tomcat 9.

Java Full Stack Developer | ***SynergisticIT*** *– Fremont, CA*

- ***Secure Auto Insurance Portal*** **Mar 2023 – Aug 2023**

Overview:
Developed a comprehensive web-based auto insurance application designed to provide users with seamless access to insurance options based on driver, vehicle, and location data. Leveraging **Java 17**, **Spring Boot**, and **Hibernate**, the application utilizes a microservices architecture to ensure scalability and flexibility. **Spring Security** with **JWT** and refresh tokens provides robust authentication and secure access control. The responsive frontend, built with **ReactJS**, **React-Bootstrap**, and **Axios**, ensures an interactive user experience. Deployed using **AWS EC2**, **Route 53**, and **S3**, the system guarantees high availability and performance, while the **CI/CD pipeline** configured with **Jenkins** streamlines automated build, testing, and deployment processes. This solution not only enables users to efficiently manage their insurance queries and bookings but also provides a secure and scalable platform for seamless service delivery.

Backend Development:

- Designed and implemented **RESTful web services** using **Spring Boot**, enhancing scalability and modularity through a microservices architecture.
- Utilized **Spring Core** features such as dependency injection, beans, and AOP for efficient development.
- Implemented secure authentication and role-based access control using **Spring Security** with **JWT** and **refresh tokens**, ensuring robust data protection.
- Employed **Spring Data JPA** to simplify data access operations, improving code clarity and maintainability.
- Used **Hibernate** as the ORM solution to streamline interactions between Java and the **MySQL** database, participating in database schema design for insurance-related data.
- Managed image storage in **AWS S3** and deployed data in a relational **Oracle database**, demonstrating proficiency in hybrid cloud and database technologies.
- Built the application with **Maven**, streamlining dependency management and project building.
- Conducted unit testing and regression testing using **Postman** and achieved smooth integration with other modules.

Frontend Development:

- Developed a responsive user interface using **ReactJS**, employing **Contexts** and **Hooks** for dynamic state management and efficient component interactions.
- Designed a visually appealing and user-friendly interface with **React-Bootstrap** and **CSS**, ensuring accessibility and responsiveness.
- Integrated **Axios** for asynchronous data retrieval, improving user experience with real-time interactions.
- Implemented custom input validation mechanisms and developed a **JWT interceptor** to handle token-based authentication and refresh expired tokens.

Deployment:

- Configured a **CI/CD pipeline** using **Jenkins**, automating build, testing, and deployment processes for streamlined delivery.
- Deployed the application on **AWS EC2** instances with **auto-scaling** capabilities to support thousands of concurrent users.
- Used **AWS Route 53** to configure domain name system (DNS) services, enabling seamless application access across devices.

Technologies Used: Java 17, Spring 5, Spring Boot, Hibernate, Spring Security, ReactJS, JWT, Axios, React-Bootstrap, MySQL, Oracle DB, Maven, Postman, AWS S3, AWS EC2, AWS Route 53, Jenkins, Git, Stripe API, iText, HTML5, CSS.

Java Full Stack Developer | *SynergisticIT* – *Fremont, CA*

- ***Shopping Cart Application*** **July 2022 – Feb 2023**

Overview:
Developed an interactive and dynamic web-based shopping cart application designed to enhance the online shopping experience for customers and streamline operational workflows for organizations. The platform enables users to browse products, manage shopping carts, apply discounts, make payments, and manage orders, with real-time notifications. For organizations, it provides tools to manage users, products, and customer retention strategies. Built with **Node.js**, **Express.js**, and **MongoDB**, the backend offers efficient handling of complex queries related to shopping preferences, orders, and notifications. The frontend, developed using **ReactJS**, **Redux**, and **Bootstrap**, provides a responsive and engaging user interface. Deployed with a focus on performance and scalability, the application ensures smooth user interactions across various devices. The system also supports secure user authentication with **JWT** tokens and real-time data synchronization with **Axios**.

Backend Development:

- Utilized the **Express** framework, incorporating middleware, lazy loading, dependency injection, and service-oriented architecture (SOA) for efficient backend development.
- Implemented **RESTful web services** for seamless communication between the frontend and backend.
- Leveraged **JWT Tokens** for secure user authentication and role-based access control.
- Adopted **Mongoose** as the ORM for effective data management, designing schemas for user profiles, products, cart details, notifications, and order history in **MongoDB**.
- Employed **ES6 async/await** for concurrent data stream handling and optimized multithreading operations.
- Applied **CORS** and static JSON middleware to manage data calls and ensure cross-origin access.
- Mounted multiple Express applications to enhance performance and used **Express. Router** for abstracting **API routes**.
- Used **Nodemon** for efficient application development with hot reloading and cross-origin deployment support.

Frontend Development:

- Built a **Single Page Application (SPA)** using **ReactJS**, **HTML5**, **Bootstrap**, and **CSS4**, ensuring a responsive and engaging user interface.
- Utilized **Redux** for state management with a publisher-subscriber model, ensuring centralized data flow through the store.
- Designed React functional components with **Hooks** for state management, caching, navigation, and improved performance.
- Used **React-Router-Dom** for frontend routing, enabling dynamic component rendering and seamless navigation.
- Applied **lazy loading** and file compression through **Webpack** to boost performance and reduce load times.
- Integrated **Axios** for efficient **API calls** and real-time data updates.
- Ensured multi-device compatibility through **Bootstrap media queries** and fluid rendering techniques.
- Enhanced development speed with features like **hot reloading**, production builds, and source maps.
- Styled components with inline **CSS** and shared **CSS** files configured via **Babel** and Webpack.

Deployment:

- Self-hosted the **MongoDB server** and **Node.js API**, utilizing **JWT** for secure token-based authentication.
- Deployed the React application with a **build-once, deploy-anywhere** approach using **bundle.js** for streamlined distribution.
- Demonstrated deployment using both **Webpack** and **Express servers**, highlighting versatility and efficiency in deployment pipelines.

Technologies Used: Node.js, Express.js, MongoDB, Mongoose, JWT, CORS, Nodemon, ES6, ReactJS, Redux, React-Redux, React-Router-Dom, Axios, Bootstrap, HTML5, CSS4, Webpack, Babel, Git, MongoDB Server, Express Server, Webpack Server.

EDUCATION

Bachelors of Science in Engineering | *Texas Tech University*